Lecture 5. Nonparametric Estimation of Basic Quantities for Right-Censored Data (Part 1)

Kaplan–Meier and Nelson–Aalen Estimators

Contents

1 Setting **1**

2 Building Up the Estimator **1**

2.1 Discrete X without censoring 1

2.2 Discrete X with right censoring 2

2.3 Continuous X 2

3 The Kaplan–Meier (Product-Limit) Estimator **2**

4 The Nelson–Aalen Estimator **4**

4.1 Alternative estimators 4

5 Worked Example (Small Sample) **4**

6 Remarks **5**

7 The 6-MP Leukaemia Trial **5**

8 Selected exercises **6**

1 Setting

X is continuous with survival S and cumulative hazard H, under **non-informative right censoring**:

$$\{(T_i, \delta_i),\ i = 1, \dots, n\}, \qquad T_i = \min(X_i, C_i), \quad \delta_i = I(X_i \le C_i).$$

Goal: estimate S and H nonparametrically, in four steps (discrete/continuous $\times$ without/with censoring).

2 Building Up the Estimator

2.1 Discrete X without censoring

Let X have support $0 < \tilde{x}_1 < \cdots < \tilde{x}_M < \infty$, $\tilde{x}_0 = 0$. By the product-limit form (Lecture 2),

$$S(x) = P(X > x) = \prod_{\tilde{x}_j \le x} \frac{S(\tilde{x}_j)}{S(\tilde{x}_{j-1})} = \prod_{\tilde{x}_j \le x} P(X > \tilde{x}_j \mid X > \tilde{x}_{j-1})$$

$$= \prod_{\tilde{x}_j \le x} P(X > \tilde{x}_j \mid X \ge \tilde{x}_j) = \prod_{\tilde{x}_j \le x} [1 - h(\tilde{x}_j)], \qquad h(\tilde{x}_j) = P(X = \tilde{x}_j \mid X \ge \tilde{x}_j).$$

(Third equality: no mass between $\tilde{x}_{j-1}$ and $\tilde{x}_j$.) For i.i.d. $X_1, \dots, X_n$, estimate

$$\hat{h}(\tilde{x}_j) = \frac{\#\{X_i = \tilde{x}_j\}}{\#\{X_i \ge \tilde{x}_j\}} = \frac{\#\text{ deaths at } \tilde{x}_j}{\#\text{ deaths at } \tilde{x}_j, \tilde{x}_{j+1}, \dots, \tilde{x}_M}, \qquad \hat{S}(x) = \prod_{\tilde{x}_j \le x} [1 - \hat{h}(\tilde{x}_j)].$$

These are the MLEs.

2.2 Discrete X with right censoring

With $T_i = \min(X_i, C_i)$, $\delta_i = I(T_i = X_i)$ and $X_i \perp\!\!\!\perp C_i$:

$$\begin{aligned} P(T_i = \tilde{x}_j, \delta_i = 1 \mid T_i \ge \tilde{x}_j) &= \frac{P(T_i = \tilde{x}_j, \delta_i = 1)}{P(T_i \ge \tilde{x}_j)} = \frac{P(X_i = \tilde{x}_j,\ C_i \ge \tilde{x}_j)}{P(X_i \ge \tilde{x}_j,\ C_i \ge \tilde{x}_j)} \\ &= \frac{P(X_i = \tilde{x}_j)P(C_i \ge \tilde{x}_j)}{P(X_i \ge \tilde{x}_j)P(C_i \ge \tilde{x}_j)} = \frac{P(X_i = \tilde{x}_j)}{P(X_i \ge \tilde{x}_j)} = h(\tilde{x}_j). \end{aligned}$$

So the hazard is identifiable from the *observed* data:

$$\hat{h}(\tilde{x}_j) = \frac{\#\{T_i = \tilde{x}_j, \delta_i = 1\}}{\#\{T_i \ge \tilde{x}_j\}}, \qquad \hat{S}(x) = \prod_{\tilde{x}_j \le x} \Big[1 - \frac{\#\{T_i = \tilde{x}_j, \delta_i = 1\}}{\#\{T_i \ge \tilde{x}_j\}}\Big].$$

Derivation: $\hat{h}$ is the MLE under censoring

Under independent censoring the likelihood factorizes over support points: of the Y_j at risk at $\tilde{x}_j$, d_j die (probability h_j each), the rest survive past it; censored subjects contribute survival to their censoring time. Thus

$$L(h_1, \dots, h_M) \propto \prod_j h_j^{d_j}(1 - h_j)^{Y_j - d_j}, \qquad \frac{\partial \log L}{\partial h_j} = \frac{d_j}{h_j} - \frac{Y_j - d_j}{1 - h_j} = 0 \ \Rightarrow\ \hat{h}_j = \frac{d_j}{Y_j}.$$

By invariance $\hat{S}(x) = \prod_{\tilde{x}_j \le x}(1 - \hat{h}_j)$ is the MLE of S.

2.3 Continuous X

Without censoring, use the empirical survival function

$$\hat{S}(t) = 1 - \mathbb{F}_n(t) = 1 - \frac{1}{n}\sum_{i=1}^{n} I(X_i \le t).$$

With censoring: the estimator should not jump at censoring times (no event there), and a censored observation counts in Y_j up to its censoring time, then leaves.

3 The Kaplan–Meier (Product-Limit) Estimator

Motivation. Kaplan–Meier multiplies conditional survival probabilities at the failure times, each estimated from the risk set.

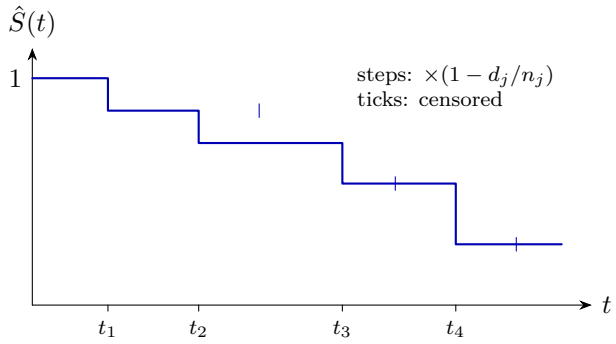


Figure. *KM drops only at failure times t_j, by $1 - d_j/n_j$; censorings (ticks) only shrink the risk set.*

Definition 5.1: Notation

- $t_1 < t_2 < \cdots < t_D$: the D distinct death times;
- $d_j = \sum_{i=1}^{n} I(T_i = t_j, \delta_i = 1)$: number of deaths at t_j;
- $Y_j = \sum_{i=1}^{n} I(T_i \geq t_j)$: number at risk just before t_j.

Theorem 5.2: Kaplan–Meier estimator

$$\hat{S}(t) = \begin{cases} 1, & t < t_1, \\ \displaystyle\prod_{t_j \leq t} \Big(1 - \frac{d_j}{Y_j}\Big), & t_1 \leq t. \end{cases}$$

Remark 5.3: Ties between deaths and censorings

A censoring tied with a death is taken to occur just *after* it, so it counts in Y_j.

Theorem 5.4: Greenwood's variance estimator

$$\hat{V}[\hat{S}(t)] = \hat{S}(t)^2 \sum_{t_j \leq t} \frac{d_j}{Y_j(Y_j - d_j)}.$$

Derivation: delta-method derivation (Cox & Oakes, 1984)

With $\hat{h}_j = d_j/Y_j$, $d_j \mid Y_j \sim \mathrm{Bin}(Y_j, h_j)$ and the $\hat{h}_j$ approximately uncorrelated,

$$\mathrm{Var}(\hat{h}_j) \approx \frac{\hat{h}_j(1 - \hat{h}_j)}{Y_j}.$$

With $\ln \hat{S}(t) = \sum_{t_j \leq t} \ln(1 - \hat{h}_j)$ and the delta method ($g(u) = \ln(1-u)$, $g'(u) = -1/(1-u)$):

$$\begin{aligned} \mathrm{Var}\{\ln \hat{S}(t)\} &\approx \sum_{t_j \leq t} \mathrm{Var}\{\ln(1 - \hat{h}_j)\} \approx \sum_{t_j \leq t} \Big(\frac{1}{1 - \hat{h}_j}\Big)^2 \mathrm{Var}(\hat{h}_j) \\ &\approx \sum_{t_j \leq t} \Big(\frac{1}{1 - \hat{h}_j}\Big)^2 \frac{\hat{h}_j(1 - \hat{h}_j)}{Y_j} = \sum_{t_j \leq t} \frac{\hat{h}_j}{Y_j(1 - \hat{h}_j)} = \sum_{t_j \leq t} \frac{d_j/Y_j}{Y_j(Y_j - d_j)/Y_j} \\ &= \sum_{t_j \leq t} \frac{d_j}{Y_j(Y_j - d_j)}. \end{aligned}$$

A second delta step: $\hat{V}[\hat{S}(t)] = \hat{S}(t)^2 \operatorname{Var}\{\ln \hat{S}(t)\}$.

4 The Nelson–Aalen Estimator

Motivation. Summing the failure rates d_j/Y_j estimates the cumulative hazard, with better small-sample behaviour.

Theorem 5.5: Nelson–Aalen estimator and Aalen's variance

$$\tilde{H}(t) = \begin{cases} 0, & t < t_1, \\ \displaystyle\sum_{t_j \le t} \frac{d_j}{Y_j}, & t_1 \le t, \end{cases} \qquad \sigma^2_{\tilde{H}}(t) = \sum_{t_j \le t} \frac{d_j}{Y_j^2}.$$

Remark 5.6: Intuition

Each death time adds the hazard increment d_j/Y_j. Treating d_j as Poisson with mean $Y_j h_j$, $\operatorname{Var}(d_j/Y_j) \approx d_j/Y_j^2$.

4.1 Alternative estimators

- Alternative for H: $\hat{H}(t) = -\log \hat{S}(t)$.
- Alternative for S: $\tilde{S}(t) = \exp\{-\tilde{H}(t)\}$ (Breslow / Fleming–Harrington).
- $1 - x \le e^{-x}$ gives $\hat{S}(t) \le \tilde{S}(t)$, i.e. $\tilde{H}(t) \le \hat{H}(t)$.
- "Product-limit": as $n \to \infty$ the number of factors grows and the product tends to a *product integral* over the support.

5 Worked Example (Small Sample)

Example 5.7: $n = 5$

Right-censored continuous X:

T_i	0.5	1	0.75	0.25	0.75
δ_i	1	1	1	0	0

Ordered: 0.25^+, 0.5, 0.75, 0.75^+, 1; at 0.75 the censored subject is at risk.

t_j	d_j	Y_j	$1 - d_j/Y_j$	$\hat{S}(t_j)$	d_j/Y_j	$\tilde{H}(t_j)$	$\tilde{S}(t_j) = e^{-\tilde{H}}$
0.5	1	4	3/4	0.750	0.250	0.250	0.779
0.75	1	3	2/3	0.500	0.333	0.583	0.558
1	1	1	0	0.000	1.000	1.583	0.205

Greenwood: $\hat{V}[\hat{S}(0.5)] = 0.75^2 \cdot \frac{1}{12} = 0.0469$, $\hat{V}[\hat{S}(0.75)] = 0.5^2(\frac{1}{12} + \frac{1}{6}) = 0.0625$; undefined at $t = 1$ ($Y_j = d_j$).

6 Remarks

- $\hat{S}$, $\tilde{H}$, $\hat{H}$ are right-continuous step functions.
- $\hat{S}$, $\tilde{S}$ are consistent for $t \le \tau$ with $P(T \ge \tau) > 0$, and asymptotically equivalent.
- Both are defined only up to $t_{\max} = \max T_i$; if the largest observation is censored, $\hat{S}$ does not reach 0. Extensions beyond $t_{\max}$:
  - Efron (1967): $\hat{S}(t) = 0$;
  - Gill (1980): $\hat{S}(t) = \hat{S}(t_{\max})$;
  - Brown, Hollander & Kowar (1974): $\hat{S}(t) = \exp\{t \log[\hat{S}(t_{\max})]/t_{\max}\}$.

7 The 6-MP Leukaemia Trial

Data. 42 children with acute leukaemia in remission, matched in pairs by remission status, randomized within pairs to 6-mercaptopurine (6-MP) or placebo, and followed to relapse or study end. Weeks to relapse, 6-MP ($^+$ = censored):

$$6,\ 6,\ 6,\ 6^+,\ 7,\ 9^+,\ 10,\ 10^+,\ 11^+,\ 13,\ 16,\ 17^+,\ 19^+,\ 20^+,\ 22,\ 23,\ 25^+,\ 32^+,\ 32^+,\ 34^+,\ 35^+.$$

Placebo (all observed): $1, 1, 2, 2, 3, 4, 4, 5, 5, 8, 8, 8, 8, 11, 11, 12, 12, 15, 17, 22, 23$.

Example 5.8: Kaplan–Meier estimate for the 6-MP group

t_j	d_j	Y_j	$\hat{S}(t) = \prod(1 - d_j/Y_j)$	$\sum \frac{d_j}{Y_j(Y_j - d_j)}$	$\hat{S}(t)^2 \sum \cdots$
6	3	21	$[1 - \frac{3}{21}] = 0.857$	$\frac{3}{21\times 18} = 0.0079$	0.0058
7	1	17	$0.857(1 - \frac{1}{17}) = 0.807$	$0.0079 + \frac{1}{17\times 16} = 0.0116$	0.0076
10	1	15	$0.807(1 - \frac{1}{15}) = 0.753$	$0.0116 + \frac{1}{15\times 14} = 0.0164$	0.0093
13	1	12	$0.753(1 - \frac{1}{12}) = 0.690$	$0.0164 + \frac{1}{12\times 11} = 0.0240$	0.0114
16	1	11	$0.690(1 - \frac{1}{11}) = 0.628$	$0.0240 + \frac{1}{11\times 10} = 0.0330$	0.0130
22	1	7	$0.628(1 - \frac{1}{7}) = 0.538$	$0.0330 + \frac{1}{7\times 6} = 0.0569$	0.0164
23	1	6	$0.538(1 - \frac{1}{6}) = 0.448$	$0.0569 + \frac{1}{6\times 5} = 0.0902$	0.0181

Censorings shrink Y_j without a jump, e.g. 9^+ between $t = 7$ ($Y = 17$) and $t = 10$ ($Y = 15$).

Example 5.9: Nelson–Aalen estimate for the 6-MP group

t	$\tilde{H}(t) = \sum d_j/Y_j$	$\sigma_H^2 = \sum d_j/Y_j^2$	SE
$0 \le t < 6$	0	0	0
$6 \le t < 7$	$3/21 = 0.1428$	$3/21^2 = 0.0068$	0.0825
$7 \le t < 10$	$0.1428 + 1/17 = 0.2017$	$0.0068 + 1/17^2 = 0.0103$	0.1015
$10 \le t < 13$	$0.2017 + 1/15 = 0.2683$	$0.0103 + 1/15^2 = 0.0147$	0.1212
$13 \le t < 16$	$0.2683 + 1/12 = 0.3517$	$0.0147 + 1/12^2 = 0.0217$	0.1473
$16 \le t < 22$	$0.3517 + 1/11 = 0.4426$	$0.0217 + 1/11^2 = 0.0299$	0.1729
$22 \le t < 23$	$0.4426 + 1/7 = 0.5854$	$0.0299 + 1/7^2 = 0.0503$	0.2243
$23 \le t < 35$	$0.5854 + 1/6 = 0.7521$	$0.0503 + 1/6^2 = 0.0781$	0.2795

Comparison. $\tilde{S} = e^{-\tilde{H}}$ lies slightly *above* KM (K&M Fig. 4.1A), most in the tail (at $t = 23$: $\hat{S} = 0.448$, $\tilde{S} = 0.471$).

8 Selected exercises

Motivation. Without censoring KM is the empirical survival function; KM redistributes censored mass to the right; Nelson–Aalen matches the moments of H at order statistics.

Exercise 5.1 (No censoring: KM and Greenwood reduce to binomial quantities)

Suppose no observation is censored ($\delta_i = 1$ for all i; ties allowed).
(a) Show that $\hat{S}(t) = 1 - \mathbb{F}_n(t)$ for all t.
(b) Show that Greenwood's formula reduces to $\hat{V}[\hat{S}(t)] = \hat{S}(t)\{1-\hat{S}(t)\}/n$ whenever $\hat{S}(t) > 0$.

Solution [Supplement].

(a)

Step 1. With $Y_{D+1} = 0$: without censoring $Y_{j+1} = Y_j - d_j$, $Y_1 = n$.

Step 2. So $1 - d_j/Y_j = Y_{j+1}/Y_j$, and for $t_k \le t < t_{k+1}$ ($t_{D+1} = \infty$)

$$\hat{S}(t) = \prod_{j=1}^{k} \frac{Y_{j+1}}{Y_j} = \frac{Y_{k+1}}{n} = \frac{\#\{i : T_i > t\}}{n} = 1 - \mathbb{F}_n(t).$$

(For $t < t_1$ both are 1.)

(b) If $\hat{S}(t) > 0$, $Y_{k+1} > 0$, and

$$\frac{d_j}{Y_j(Y_j - d_j)} = \frac{Y_j - Y_{j+1}}{Y_j Y_{j+1}} = \frac{1}{Y_{j+1}} - \frac{1}{Y_j}, \qquad \sum_{j=1}^{k} \frac{d_j}{Y_j(Y_j - d_j)} = \frac{1}{Y_{k+1}} - \frac{1}{n}.$$

With $\hat{S} = Y_{k+1}/n$,

$$\hat{V}[\hat{S}(t)] = \frac{Y_{k+1}^2}{n^2}\Big(\frac{1}{Y_{k+1}} - \frac{1}{n}\Big) = \frac{1}{n} \cdot \frac{Y_{k+1}}{n}\Big(1 - \frac{Y_{k+1}}{n}\Big) = \frac{\hat{S}(t)\{1 - \hat{S}(t)\}}{n},$$

the binomial variance estimate. ■

Exercise 5.2 (Self-consistency of the Kaplan–Meier estimator)

Let t be such that $\hat{S}(t) > 0$. Show that

$$\hat{S}(t) = \frac{1}{n}\Big[\#\{i : T_i > t\} + \sum_{i:\, \delta_i = 0,\ T_i \le t} \frac{\hat{S}(t)}{\hat{S}(T_i)}\Big].$$

I.e. each subject censored at $T_i \le t$ counts as alive at t with probability $\hat{P}(X > t \mid X > T_i)$.

Solution [Supplement].

Step 1. For $s \le t$ let $R(s) = \#\{i : T_i > s\}$, $A(s) = \sum_{i:\delta_i=0, T_i \le s} 1/\hat{S}(T_i)$ (well defined, $\hat{S}(s) \ge \hat{S}(t) > 0$), and

$$\varphi(s) = \frac{R(s)}{\hat{S}(s)} + A(s).$$

The claim is $\varphi(t) = n$.

Step 2. R, $\hat{S}$, A change only at observed times; for $s < \min_i T_i$, $\varphi(s) = n$.

Step 3. At an observed $s \le t$ with $Y = R(s-)$ at risk, d deaths and c censorings: $R(s) = Y - d - c$, $A(s) = A(s-) + c/\hat{S}(s)$, $\hat{S}(s) = \hat{S}(s-)(1 - d/Y)$. Hence

$$\varphi(s) - \varphi(s-) = \frac{Y - d - c}{\hat{S}(s)} + \frac{c}{\hat{S}(s)} - \frac{Y}{\hat{S}(s-)} = \frac{(Y-d) - Y(1-d/Y)}{\hat{S}(s)} = 0.$$

(The tie convention puts the c censored subjects in Y.)

Step 4. So $\varphi \equiv n$ on $(-\infty, t]$.

Efron (1967): this "redistribute-to-the-right" property characterizes $\hat{S}$; it underlies Turnbull's algorithm (Lecture 8). ■

Exercise 5.3 (Nelson–Aalen at the order statistics)

Let $X_1, \dots, X_n$ be i.i.d. with continuous F, no censoring, and order statistics $X_{(1)} < \cdots < X_{(n)}$.
(a) Show that $\tilde{H}(X_{(k)}) = \sum_{i=1}^{k} \frac{1}{n-i+1}$ and $\sigma_H^2(X_{(k)}) = \sum_{i=1}^{k} \frac{1}{(n-i+1)^2}$.
(b) Show that $\mathrm{E}\{H(X_{(k)})\} = \tilde{H}(X_{(k)})$ and $\mathrm{Var}\{H(X_{(k)})\} = \sigma_H^2(X_{(k)})$.

Solution [Supplement].

(a) No ties a.s., so $d_i = 1$, $Y_i = n - i + 1$ at $X_{(i)}$; use Theorem 5.5.

(b)

Step 1. $S(X) \sim U(0,1)$, so $P\{H(X) > u\} = P\{S(X) < e^{-u}\} = e^{-u}$: $E_i = H(X_i)$ are i.i.d. Exp(1).

Step 2. H is nondecreasing: $H(X_{(k)}) = E_{(k)}$.

Step 3. $(E_{(1)}, \dots, E_{(n)})$ has density $n! \exp(-\sum_i e_i)$ on $0 < e_1 < \cdots < e_n$. The spacings $D_i = E_{(i)} - E_{(i-1)}$ $(E_{(0)} = 0)$ have unit Jacobian and $\sum_i e_i = \sum_i (n-i+1) d_i$, so their density is

$$\prod_{i=1}^{n} (n-i+1)\, e^{-(n-i+1)d_i}, \qquad d_i > 0,$$

i.e. independent $D_i \sim \mathrm{Exp}(n-i+1)$.

Step 4. $E_{(k)} = \sum_{i \le k} D_i$ has mean $\sum_{i \le k} (n-i+1)^{-1}$ and variance $\sum_{i \le k} (n-i+1)^{-2}$; compare (a). ■